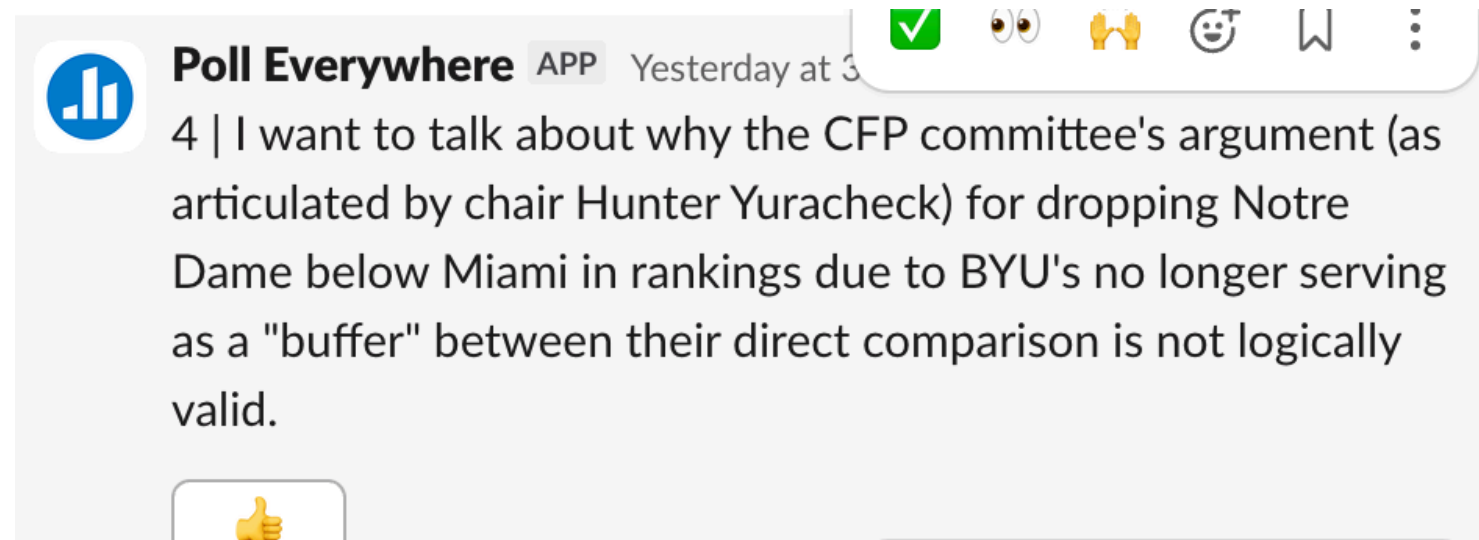


Our last
class.

As you know, the topics for our last class day were up to you. Here is one suggestion that caught my attention:



Was the College Football Playoff committee's decision to exclude Notre Dame competent, or incompetent?

Was the College Football Playoff committee's decision to exclude Notre Dame honest, or corrupt?

Logic tells us that there are four exclusive and exhaustive possibilities.

The decision
was
competent
and honest.

The decision
was
competent
and corrupt.

The decision
was
incompetent
and honest.

The decision
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and corrupt.

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The decision
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and corrupt.

Some find visual evidence of incompetence.



But, as you know, in philosophy we rely on arguments.

The decision
was
competent
and honest.

Here is an argument against the first possibility.

1. If the committee's decision was competent and honest, then the committee did not rely on absurd principles in making its decision.
 2. The committee relied on absurd principles in making its decision.
-
- C. The committee's decision was not competent and honest. (1,2)

The argument is valid, and the first premise is obvious. It remains therefore to demonstrate the second premise.

The committee's task is to rank the country's football teams from best to worst, based on their performance. This is a general kind of task. One might also seek to similarly rank the best universities, or best philosophy departments, or best cars, or whatever.

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The committee's task is to rank the country's football teams from best to worst, based on their performance. This is a general kind of task. One might also seek to similarly rank the best universities, or best philosophy departments, or best cars, or whatever.

One may rely on various principles in constructing rankings of this kind, and the appropriate principles will of course depend on the subject matter of the ranking. Nonetheless we can isolate one general kind of principle which one might use, which I will call a **magic touching rule**.

A magic touching rule is a rule which says that when two of the ranked items are 'touching' — i.e., adjacent in the rankings — then we look to a special extra condition — this is the magic part — to see whether the two touching items in the ranking should have their spots flipped.

The committee employs a rule of this kind.

The decision
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The committee employs a rule of this kind.

Magic touching rule

If two teams in the ranking are touching and played head to head and the lower ranked team defeated the higher ranked team then the teams' rankings are flipped.

It is easy to argue that magic touching rules are, in general, dumb. Either the extra condition (in this case the head to head result) is relevant to the ranking or it is not. If it is, it should be incorporated into the original ranking, in which case the flip is 'double counting' the relevance of the extra condition. If it is not relevant, it should also not be relevant in the special case of touching.

Further absurdities follow from the fact that the extra condition (winning a head to head matchup) is not a transitive relation.

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Further absurdities follow from the fact that the extra condition (winning a head to head matchup) is not a transitive relation.

Suppose that we have a case in which Team A beat Team B, Team B beat Team C, and Team C beat Team A. Suppose further that we have arrived at the following initial rankings:

Team A

Team C

Team B

We obviously cannot stick with these rankings, since A and C are touching, and C beat A. But C and B are also touching, and B beat C. What should we do?

The decision
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competent
and honest.

Team A

Team C

Team B

We obviously cannot stick with these rankings, since A and C are touching, and C beat A. But C and B are also touching, and B beat C. What should we do?

Suppose we apply magic touching to the first pair. Then we get these rankings. Phew; no further flips required.

Team C

Team A

Team B

Suppose instead we apply magic touching to the second pair. Then we get these rankings.

Team A

Team B

Team C

Results depend on arbitrary choices about the order of rule application.

The decision
was
competent
and honest.

Here is a second argument against the first option.

1. If the committee's decision was competent and honest, then the committee did not inconsistently apply its own principles.
 2. The committee inconsistently applied its own principles in making its decision.
-
- C. The committee's decision was not competent and honest. (1,2)

Here is one rule the committee sometimes appears to endorse:

13th data point rule
Teams cannot be moved down in
the rankings due to their
performance in conference
championship games.

The decision
was
competent
and honest.

13th data point rule
Teams cannot be moved down in
the rankings due to their
performance in conference
championship games.

Behold the 5th playoff
rankings:

9. Alabama
10. Notre Dame
11. BYU
12. Miami

After Alabama and BYU lost conference
championship games by comparable margins with
ND and Miami idle, one would expect, given the
above rule, that the rankings would remain
unchanged.

However, the committee ignored the 13th data
point rule in the case of BYU but not Alabama;
BYU's drop then enabled magic touching to
come into play.

I conclude that the committee's decision was not both competent and honest. So three possibilities remain.

The decision
was
competent
and corrupt.

The decision
was
incompetent
and honest.

The decision
was
incompetent
and corrupt.

We can argue against the first as follows:

1. If the committee's decision was competent and corrupt, then the committee did not rely on obviously absurd principles and did not obviously inconsistently apply its own principles.
2. The committee relied on obviously absurd principles in making its decision and obviously inconsistently applied its own principles.

C. The committee's decision was not competent and corrupt. (1,2)

If the committee were competent, and corrupt, we would not find obvious evidence of this; and yet we do. So, if they are corrupt, they are not competently so.

The decision
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incompetent
and honest.

The decision
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and corrupt.

Now we face our most difficult question. Was the committee incompetent and honest or incompetent and corrupt? In our current state of evidence, we do not have an argument which decisively settles this question.

However, we have already discussed a mechanism for arriving at probabilistic beliefs in such cases: Bayes' theorem.

Recall that in order to determine the probability of some hypotheses given the evidence, we need two pieces of information: the prior probabilities of the theories, and the probability of the evidence given the theories.

Given the composition of the committee, some are antecedently inclined to think them corrupt. But, being fair-minded, let's say that the prior probabilities that the committee was incompetent and corruptly biased toward conferences and incompetent and not so corruptly biased are each 0.5.

Given the composition of the committee, some are antecedently inclined to think them corrupt. But, being fair-minded, let's say that the prior probabilities that the committee was incompetent and corruptly biased toward conferences and incompetent and not so corruptly biased are each 0.5.

$$P(\text{CORRUPT}) = 0.5$$

$$P(\text{HONEST}) = 0.5$$

Our evidence (e) will be the committee's decisions from the 4th playoff rankings onwards.

As we will see, there is only a narrow path for these decisions which would have allowed Notre Dame to be excluded. Despite the narrowness of the path and the committee's general incompetence, the probability of a series of Notre Dame-excluding decisions conditional on CORRUPT is high. (Even a relatively unintelligent committee would be able to identify the Notre Dame-excluding paths.) Choice of numbers are somewhat arbitrary, but the following appears to be a conservative choice:

$$P(e \mid \text{CORRUPT}) = 0.8$$

$$P(\text{CORRUPT}) = 0.5$$

$$P(e \mid \text{CORRUPT}) = 0.8$$

$$P(\text{HONEST}) = 0.5$$

The key remaining question is then: what is the probability of those decisions conditional on HONEST? To that question I now turn.

This is the relevant portion of the week 4 rankings.

8. Oklahoma
9. Notre Dame
10. Alabama
11. BYU
12. Miami

In the subsequent week, all five teams played weak opponents; Notre Dame, BYU, and Miami won easily, and Oklahoma and Alabama won narrowly.

The fact that all teams won made (according to betting sites) the status quo most likely.

But the following outcomes were also possible, given the narrow wins by Oklahoma and Alabama.

8. Oklahoma
9. Notre Dame
10. Alabama
11. BYU
12. Miami

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The committee surprised the betting markets:

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Inspection reveals that of these five possible outcomes only one permitted the possibility of eventual magic touching between Notre Dame and Miami. So the question is: what were the odds that an honest committee would settle on this, out of the five?

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The markets took this to be the least likely of the five outcomes, so, if we trust the markets, it should be some number less than 0.2. However, the markets presumably were pricing in the possibility of CORRUPT, which suggests that it should be a number significantly lower than this.

A generous estimate would then appear to be a probability of 0.15. Here as elsewhere, if you think that the correct probability is lower or higher, you can pick your own number and generate your own result.

We now turn to the last week, which began with the rankings in yellow, and all teams idle other than Alabama and BYU, who lost by comparable scores.

8. Oklahoma
9. Alabama
10. Notre Dame
11. BYU
12. Miami

Given the previous discussion of the 13th data point rule, the markets again expected (overwhelmingly) the status quo.

The committee had five live options:

- (1) Status quo
- (2) Drop both Alabama and BYU one slot
- (3) Drop Alabama but not BYU one slot
- (4) Drop BYU but not Alabama by one slot
- (5) Drop BYU but not Alabama by one slot, and then employ magic touching to flip Notre Dame and Miami

Obviously only (5) excludes Notre Dame, and I will again follow betting markets in setting the probability of this option at 0.1.

As above, this is generous, given that the markets observed the bizarre week 4 decision and hence must have been taking CORRUPT seriously by now.

$$P(\text{CORRUPT}) = 0.5$$

$$P(e \mid \text{CORRUPT}) = 0.8$$

$$P(\text{HONEST}) = 0.5$$

Because Notre Dame's exclusion required both events to obtain, and given that the two choices of the committee would presumably be independent given HONEST, we can set the probability of both happening in the absence of corruption to the product of their individual probabilities. So:

$$P(e \mid \text{HONEST}) = 0.015$$

Recall Bayes' theorem:

$$\text{Bayes' theorem}$$
$$P(h \mid e) = \frac{P(h) * P(e \mid h)}{P(h) * P(e \mid h) + P(\neg h) * P(e \mid \neg h)}$$

It is now just a matter of arithmetic to compute the probabilities of our two theories.

$$P(\text{CORRUPT}) = 0.5$$

$$P(\text{HONEST}) = 0.5$$

$$P(e \mid \text{CORRUPT}) = 0.8$$

$$P(e \mid \text{HONEST}) = 0.015$$

It is now just a matter of arithmetic to compute the probabilities of our two theories.

$$P(\text{CORRUPT} \mid e) = 0.982$$

$$P(\text{HONEST} \mid e) = 0.018$$

I end this completely scientific demonstration with the conclusion that there is a 98.2% chance that the committee was both incompetent and corrupt.

The other most requested topics were all about my view on the various questions we have talked about this semester. So, ask me anything (within reason)!